Computer Vision for Internet of Things

Edge-Based Visual Intelligence and Object Detection

From Cloud Vision to On-Device AI (2023-2025)

Review Paper

Based on Analysis of Recent Publications

December 10, 2025

Abstract

Computer vision has become a critical component of IoT systems, enabling intelligent visual perception across smart cities, industrial automation, autonomous vehicles, and surveillance applications. This paper presents a comprehensive review of computer vision technologies for IoT deployments, examining the transition from cloud-based processing to edge and on-device visual intelligence. We analyze: (1) deep learning architectures for IoT vision including YOLO, MobileNet, and EfficientDet; (2) edge deployment strategies and hardware accelerators; (3) low-power vision systems like HyperCam for always-on IoT cameras; (4) benchmarking of object detection models on edge devices; (5) V2X cooperative perception for autonomous vehicles. Our analysis reveals that optimized models achieve 20+ FPS on embedded platforms while maintaining mAP above 0.7. HyperCam demonstrates low-power onboard vision consuming milliwatts for continuous operation. We identify key challenges including accuracy-latency trade-offs, handling diverse lighting conditions, and multi-camera coordination. The paper examines applications in traffic detection, smart city surveillance, and industrial inspection. Future directions point toward vision-language models for IoT, neuromorphic vision sensors, and collaborative edge vision systems.

Keywords: Computer Vision, IoT, Object Detection, Edge AI, YOLO, MobileNet, Smart Camera, Embedded Vision, V2X, Visual Intelligence

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1 Introduction

1.1 Visual IoT: The Rise of Smart Cameras

The proliferation of IoT cameras has created unprecedented opportunities for visual intelligence:

- Over 1 billion surveillance cameras deployed globally
- Smart cameras in retail, manufacturing, transportation
- Autonomous vehicles with multiple vision sensors
- Drone-based inspection and monitoring systems

1.2 From Cloud to Edge Vision

Traditional cloud-based vision processing faces limitations:

- **Bandwidth**: Transmitting video streams is expensive
- **Latency**: Round-trip delays unacceptable for real-time applications
- **Privacy**: Sensitive video data exposure concerns
- **Reliability**: Dependence on network connectivity

Edge-based computer vision addresses these challenges by processing visual data locally.

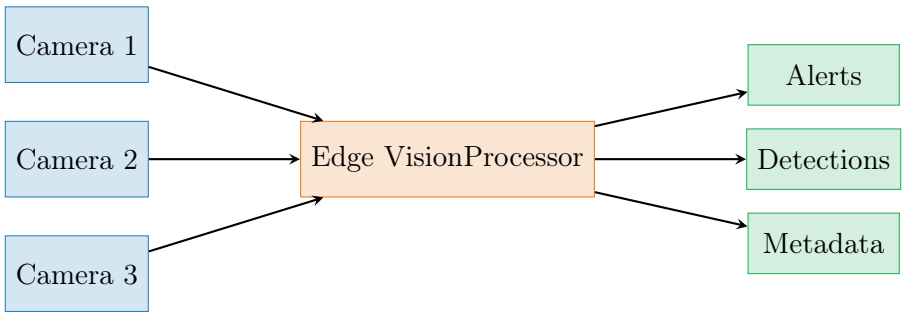


Figure 1: Edge Vision Architecture for IoT

2 Deep Learning for IoT Vision

2.1 Object Detection Architectures

Modern object detection models optimized for IoT:

Table 1: Object Detection Models for Edge Deployment

Model	mAP	FPS (Jetson)	Size	Power
YOLOv5n	0.45	45	3.9 MB	5W
YOLOv8s	0.61	30	22 MB	10W
MobileNet-SSD	0.55	25	17 MB	5W
EfficientDet-D0	0.52	20	15 MB	8W
YOLO-NAS-S	0.65	35	19 MB	8W

2.2 Model Optimization for Edge

Techniques for deploying vision models on IoT devices:

Edge Optimization Techniques

- **Quantization**: INT8/FP16 reduces memory 2-4x
- **Pruning**: Remove redundant connections
- **Knowledge Distillation**: Train smaller student models
- **Neural Architecture Search**: Optimize for target hardware
- **TensorRT/ONNX**: Hardware-specific compilation

2.3 Benchmarking on Edge Devices

Comprehensive benchmarking reveals performance across platforms:

Table 2: Vision Model Performance on Edge Hardware

Platform	YOLOv5n FPS	Power	Cost
NVIDIA Jetson Nano	25	10W	$99
Raspberry Pi 4	8	5W	$55
Google Coral	30	2W	$60
Intel NCS2	15	1W	$69
Jetson Xavier NX	80	15W	$399

3 Low-Power Vision Systems

3.1 HyperCam: Ultra-Low-Power IoT Vision

HyperCam introduces low-power onboard computer vision for IoT cameras:

- Milliwatt-level power consumption
- Always-on visual sensing capability
- Event-driven processing to minimize energy

- Hierarchical processing: simple on-device, complex off-device

3.2 Event-Based Vision

Neuromorphic vision sensors offer advantages:

- Microsecond temporal resolution
- High dynamic range (140dB vs 60dB conventional)
- Low power through sparse, event-driven output
- Ideal for motion detection in IoT applications

3.3 Duty-Cycled Vision

Energy-efficient approaches for battery-powered cameras:

1. Wake-on-motion using PIR sensors
2. Adaptive frame rate based on scene activity
3. Selective region processing
4. Hierarchical wake-up strategies

4 Smart City Vision Applications

4.1 Traffic Detection and Analysis

AI-driven traffic accident detection leveraging:

- Multi-camera fusion for comprehensive coverage
- Real-time vehicle and pedestrian tracking
- Anomaly detection for accident identification
- Integration with traffic management systems

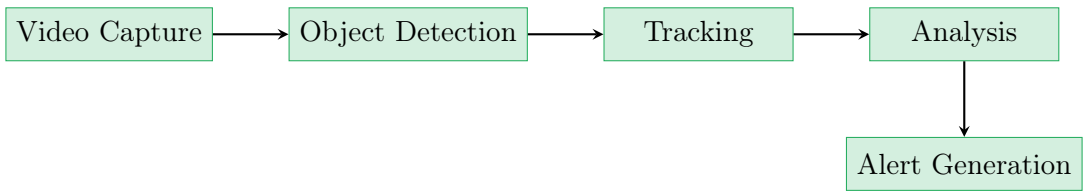


Figure 2: Traffic Vision Pipeline

4.2 Surveillance and Security

Smart camera capabilities:

- Person detection and re-identification
- Intrusion detection in restricted areas
- Crowd density estimation
- Abandoned object detection

5 V2X Cooperative Perception

5.1 Vehicle-to-Everything Vision

V2X enables cooperative perception for autonomous vehicles:

V2X Vision Benefits

- Extended perception range beyond line-of-sight
- Occlusion handling through multi-vehicle fusion
- Reduced sensor redundancy per vehicle
- Enhanced safety through information sharing

5.2 Vision-Language Models for V2X

V2X-VLM introduces end-to-end cooperative driving through:

- Large Vision-Language Models for scene understanding
- Natural language reasoning about driving scenarios
- Multi-agent communication and coordination
- Explainable decision-making for autonomous systems

5.3 Cooperative Perception Challenges

Key challenges in V2X vision:

- Communication latency and bandwidth constraints
- Sensor calibration across vehicles
- Data fusion from heterogeneous sources
- Privacy and security of shared perception data

6 Implementation Considerations

6.1 Hardware Selection

Table 3: Vision Hardware Selection Guide

Use Case	Recommended HW	Key Factor
Always-on detection	Coral/NCS2	Power efficiency
Real-time tracking	Jetson Nano/NX	FPS performance
Multi-camera system	Jetson Xavier	Throughput
Battery-powered	Custom ASIC	Ultra-low power

6.2 Camera Selection

Important camera specifications for IoT vision:

- Resolution vs. processing requirements trade-off
- Frame rate for motion applications
- Low-light performance (IR, wide aperture)
- Field of view for coverage requirements
- Interface (MIPI CSI, USB, GigE)

6.3 Software Frameworks

- **OpenCV**: Comprehensive computer vision library
- **TensorRT**: NVIDIA inference optimization
- **OpenVINO**: Intel edge inference toolkit
- **ONNX Runtime**: Cross-platform inference
- **DeepStream**: NVIDIA video analytics SDK

7 Challenges and Solutions

7.1 Accuracy-Latency Trade-offs

Balancing detection quality with real-time requirements:

- Model scaling (width, depth, resolution)
- Adaptive inference based on scene complexity
- Cascaded detection (fast screening + detailed analysis)

7.2 Environmental Robustness

Handling diverse conditions:

- Low-light and night vision challenges
- Weather effects (rain, fog, glare)
- Domain adaptation for new environments
- Data augmentation for robustness

7.3 Multi-Camera Coordination

Challenges in distributed vision systems:

- Temporal synchronization across cameras
- Spatial calibration and alignment
- Object handoff between camera views
- Bandwidth management for multi-stream processing

8 Future Directions

1. **Vision-Language IoT**: Integrating VLMs for natural language visual queries
2. **Neuromorphic Vision**: Event cameras for ultra-efficient visual sensing
3. **Collaborative Edge Vision**: Distributed processing across camera networks
4. **Self-Supervised Learning**: Reducing labeled data requirements for IoT vision
5. **Privacy-Preserving Vision**: On-device processing without raw video transmission

9 Conclusion

Computer vision has become integral to IoT systems, enabling intelligent visual perception across diverse applications. The transition from cloud to edge processing has made real-time vision feasible on embedded platforms, with optimized models achieving 20+ FPS while maintaining high accuracy. Low-power systems like HyperCam enable always-on visual sensing for battery-powered IoT devices. V2X cooperative perception extends visual intelligence to autonomous vehicle networks. Key challenges remain in balancing accuracy with efficiency, handling environmental variations, and coordinating multi-camera systems. Future developments in vision-language models and neuromorphic sensors will further expand the capabilities of IoT vision systems.

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